Isentropic Analysis

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Introduction

Famous Quotes on Isentropic Analysis

"It can hardly be doubted that the isentropic charts represent the true motion of the air more faithfully by far than synoptic charts for any fixed level of the free atmosphere"

-Rossby et. al, "Isentropic Analysis", 1937

"The isentropic chart serves a twofold purpose as (1) a hydrodynamic, and (2) a thermodynamic chart - hydrodynamic in the sense that "tongues" of maximum water-vapor content serve as identifying indicators of the flow patterna and lateral mixing; and thermodynamic with respect to indications of adiabatic changes in the air, including water vapor, as it flows along sloping or broadly undulating isentropic surfaces."

-H.R. Byers, "On the Thermodynamic Interpretation of Isentropic Charts", 1938

Before the switch to pressure coordinates prompted by aviation and WWII in the early 1940s, meteorologists regularly performed isentropic analysis as part of their analysis and diagnosis process. Isentropic analysis, as described by Rossby and many others, confers certain advantages in the forecasting process. Some utilities of isentropic analysis are outlined below:

- Diagnose and visualize vertical motion - through advection of pressure using system-relative flow
- Diagnose frontal boundaries and airmass types in the horizontal and vertical, and in a conveyor belts framework
- Depict frontogenetical and jet streak circulations
- Depict 3D transport of moisture
- Use isentropic potential vorticity (IPV) to depict the dynamic tropopause and tropopause folds
- Diagnose upper-level and dry static stability
- Diagnose symmetric instabilities

This lab will focus on the first two utilities, and will link the analyzed airmasses and frontal structures to the more standard way you have viewed weather structures and features in pressure coordinates.

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